UNIVERSITY OF WESTMINSTER

5COSC020W DATABASE SYSTEMS
2026-2027 Week 5 Coursework Support Tutorial

As a priority, concentrate on questions 1, 2.1 ahead of the Intermediate submission deadline on Thursday 22 October 2026 at 13:00:00. If you have completed Part A, move to Part B.

Tutorial for Coursework Development – Coursework Part A

1. COURSEWORK QUESTION 1 – Evaluation of the AI-generated EERD for Westlivingster

(a) On the AI-generated EERD, identify **2 weaknesses** related to conceptual **modelling conventions**.

- Issues related to the UML notation used in this module, and elements that do not belong in a conceptual model.
- Consider the ***notations*** of the AI-generated EERD: diagrammatic representations and notations.
- Consider the ***elements*** that have been added or are missing on this AI-generated EERD

(b) On the AI-generated EERD, identify **3 weaknesses** related to conceptual **semantics**.

- Issues related to how the diagram fails to capture the meaning of Westlivingster's data correctly: its entities, specialisations, relationships and multiplicities, or attributes. Each must refer to the brief.
- Consider the ***semantics*** of the generated EERD: entities, specialisations, relationships, multiplicities, attributes and primary keys.

2.1. COURSEWORK QUESTION 2.1: The Conceptual EERD for Westlivingster

Carefully consider the provided **Project Brief** and the **Executive Summary of Answers to the Business Questions (ESABQ).**

(a) Identify the relevant **entities** for the data needs of the business**.**

- Essential business concepts for Westlivingster and on which data need to be stored.
- Make sure that you capture the entities with a ***physical existence*** (e.g. Book, Book_Copy, etc. for a library) as well as those that are ***more abstract*** and therefore purely conceptual (e.g. Reservation, Loan, etc. for a library).

(b) Identify possible **specialisations** of the entities**.**

- Sub-types of specific entities for which different types can be identified.
- Make sure you identify the correct ***constraints*** to describe the participation and overlap of the sub-types in the specialisation.
  {Optional, Or}; {Optional, And}; {Mandatory, Or} or {Mandatory, And}.
- Make sure you clearly distinguish which attributes and relationships are ***inherited*** from those that are ***specific*** to the specialisations.

(c) Identify the **relationships** between entities.

- Conceptual associations between instances of entities.
- Make sure you capture relationships that are ***highly meaningful*** for the business and for which instances of entities must be connected.
- Make sure you avoid ***redundant relationships***.
- Verify whether a ***ternary*** or a ***quaternary*** relationship is needed.

(d) Identify the **multiplicities** for the relationships between entities.

- Descriptors for how an entity is related to another entity.
- Make sure that for each relationship, you identify the ***participation*** and the ***cardinality***.

(e) Identify the **attributes** and **Primary Keys** for each entity.

- The data items that need to be recorded for an entity.
- Make sure you select the most appropriate attributes to act a Primary Key i.e. a unique identifier that is irreducible.

2.2. COURSEWORK QUESTION 2.2: The Simplified data dictionary

(a) In the provided table, list the relevant **entities** for the data needs of the business, their **specialisations** and the **specialisation constraints**.

- **Entities** as essential business concepts for Westlivingster and on which data need to be stored.
- **Specialisations** as sub-types of specific entities for which different types can be identified.
- **Specialisation Constraints** to describe the participation and overlap of the sub-types in the specialisation. {Optional, Or}; {Optional, And}; {Mandatory, Or} or {Mandatory, And}

(b) In the provided table, list the relevant **relationships** between the entities with their multiplicities.

- **Relationships** that are conceptually mean.
- **Multiplicities** to define the **participation** (minimum) and the **cardinality** (maximum).

Tutorial for Coursework Development – Coursework Part B

Carefully consider the provided **Conceptual EERD for Tourmato.**

3.1. COURSEWORK QUESTION 3.1: LOGICAL MAPPING.

a. Identify and categorise the **relationships** available on the provided **Conceptual EERD for Tourmato** and associate them to the **Mapping Rules** (see lecture 03)**.**

i. Identify the specialisations (rules 7-10).
ii. Identify the one-to-one relationships mandatory on both sides (rule 2).
iii. Identify the many-to-many relationships and the ternary relationships (rules 5-6).
iv. Identify the one-to-many relationships (rule 1), one-to-one relationships optional on one side (rule 3), one-to-one relationships optional on both sides (rule 4).

b. Apply the **mapping rules** to resolve each relationship and to create your new **logical ERD for Tourmato** on draw.io

i. Use rules 7-10 for the specialisations.
ii. Use rule 2 for the one-to-one relationships mandatory on both sides.

iii. Use rule 5-6 for the many-to-many relationships and the ternary relationships.
iv. Use rule 1 for the one-to-many relationships.
v. Use rule 3 for the one-to-one relationships optional on one side.
vi. Use rule 4 for the one-to-one relationships optional on both sides.

3.2. COURSEWORK QUESTION 3.2: MAPPING LOG.

In the table provided:

a. Justify how you mapped the **specialisations:** explain how you used rules 7-10

b. Justify how you mapped the **one-to-one relationships mandatory on both sides:** explain how you use rule 2.

c. Justify how you mapped the **many-to-many relationships** and **the ternary relationships:** explain how you use rule 5 & 6.

d. Justify how you mapped the **one-to-many relationships, one-to-one relationships optional on one side** and **one-to-one relationships optional on both sides:** explain how you use rule 1, 3 and 4.